

Enumeration completeness recheck against the amended class: all enumerated readings are admissible, with the decagonal star extremal at the endpoint

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

The U1 candidate theorem quantifies over the H-ADM-COH-amended class; B1's evidence quantifies over the enumerated ensembles. For the two to compose without a gap, every enumerated pattern must BE an amended-class member (else the races would be evidence about excluded objects). This note performs that recheck with exact geometry, and extracts the extremal finding. Scope: single-shell readings of the enumerated set + the Math437 gallery; the two-shell ensembles inherit per-shell (their within-shell stars are from the same list).

2. The separation table (exact geometry)

The amended-class criterion: pairwise READING-AXIS separations $\geq \theta_{\min}(\hat{r}) = \sqrt{\hat{r}}/(2q_0^2\sqrt{C})$, with $\theta_{\min} = 0.6027/0.6131/0.6272$ rad at $I = 4 \times 10^{-4}/10^{-3}/2 \times 10^{-3}$ ($\hat{r} = 0.3110/0.3206/0.33675$).

Pattern (n axes)	min axis angle (exact)	rad	in-class margin (anchor / endpoint)
LAM (1)	— (single axis)	—	trivially in-class
square (2)	90°	1.5708	$\times 2.61$ / $\times 2.50$
SC{100} (3)	90°	1.5708	$\times 2.61$ / $\times 2.50$
HEX (3)	60°	1.0472	$\times 1.74$ / $\times 1.67$
FCC (4)	$\arccos(1/3) \rightarrow \min 70.53^\circ$	1.2310	$\times 2.04$ / $\times 1.96$
BCC (6)	60°	1.0472	$\times 1.74$ / $\times 1.67$
icosahedral (6)	$\arctan 2 = 63.43^\circ$	1.1071	$\times 1.84$ / $\times 1.77$
decagonal QC (5)	36°	0.62832	$\times 1.042$ / $\times 1.0018$

ALL patterns are amended-class members at all three anchor intensities: the races and the class theorem compose with no coverage gap. The two-shell ensembles (Math432) use within-shell stars from this list and inherit the verdict per shell (cross-shell separations live in the radial direction and are not the angular criterion's object).

3. The extremal finding and the pre-registered stress test

The decagonal star is the EXTREMAL in-class pattern: margin $\times 1.042$ at the anchor, $\times 1.0018$ at the endpoint — it sits essentially AT the class boundary at $I = 2 \times 10^{-3}$ (0.62832 vs 0.62716 rad, 0.18%). Consequences:

- **Stress test (pre-registered)**: the decagonal star at the endpoint intensity is the canonical worst-geometry probe for the STEP-5B falsifier set — any future tightening of the chain should be exercised on it first. (Energetically it is a WEAK competitor — the Math437 gallery records ZERO non-trivial 4-wave resonances for it — but for the beyond-layer machinery it is the packing-extremal member.)
- **Boundary softness**: the class boundary is not a cliff — the AddC lemma bounds any boundary-crossing restructuring by $c_{\text{ind}} I^2 = \text{margin}/\times 33$ or better, so the decagonal’s near-boundary position creates no discontinuity threat to the U1 theorem.
- **Sea reclassification, formalized**: k -fold planar stars with axis spacing $\pi/k < \theta_{\min}$ ($k \geq 6$ in-plane axes, i.e. 12-fold symmetric and denser; at intensities slightly above the endpoint even the 10-fold) are NOT class members — their sub-cutoff structure belongs to the Gaussian sea. This formalizes why dense angular stars never surfaced as estimator threats in the races: they are not independent-reading competitors at the coherence resolution.

4. Numerical verification

The table’s angle values are EXACT geometry (cube-diagonal $\arccos(1/3)$; icosahedral $\arctan 2$; planar k -fold π/k); $\theta_{\min}(\hat{r})$ values are closed-form on certified constants ($q_0^2 = 0.46264$, $\hat{r}(I) = r_R + 2\lambda'I$). NO machine asserts were executed this session (sandbox shell unavailable); the angle table + margins are registered for the shared follow-up script (planned addition to codes/vacuum/third_cumulant_assessment.py or a sibling: exact-geometry asserts to 10^{-12} , margin table to 10^{-6} , JSON artefact under claims/B1-RH-ENUM/runs/).

Quantitative sanity check (mandatory): dimensional — $\theta_{\min} = \sqrt{\hat{r}}/(2q_0^2\sqrt{C})$ is a pure angle, since $[\hat{r}] = [Cq_0^4]$ makes the ratio dimensionless (the AddB check); consistency — the anchor $\theta_{\min} = 0.6027$ reproduces AddB’s 0.603, and $n_{\text{pack}} = 16/\theta_{\min}^2 = 44.0/42.6/40.7$ reproduces the AddC “44/43/41” to rounding; limit case — as I grows, $\theta_{\min}$ grows and the first casualty of the class is the densest star (decagonal), exactly the monotonic behavior a resolution cutoff must have; cross-check — the decagonal endpoint margin $0.62832/0.62716 = 1.00185$ is consistent with the quoted 0.18%.

5. Devil’s-advocate (three objections; CLAUDE.md 6.3)

- (α) “If the decagonal is only 0.18% inside the class at the endpoint, a tiny upward revision of $\theta_{\min}$ (e.g. the criterion-band ambiguity $c_\theta \in [1, \pi]$) EXPELS it — and then the B1 races would include an out-of-class pattern, breaking the composition.” DISMISSED with the composition’s actual logic: the U1 theorem quantifies over the amended class; if a raced pattern exits the class under a stricter criterion, the race for it becomes evidence about a sea-reclassified configuration — harmless surplus, not a gap (the class theorem never RELIED on the races; U1 made them corroboration). The composition requires class members to be COVERED by the theorem (they are), not that every raced object stay a member.
- (β) “The two-shell inheritance is hand-waved: cross-shell angular proximity (a reading on shell 1 nearly parallel to a reading on shell 2) could violate the coherence criterion in a way the per-shell table misses.” VALID-with-mitigation: the coherence criterion of AddB is built from the TRANSVERSE beat scale on a shell; for cross-shell pairs the beat acquires the RADIAL separation $\Delta q \geq$ the shell gap, which exceeds $1/\xi$ by construction in the two-shell ensembles (Lemma J’s $\times 1.70$ floor operates at exactly this scale) — nearly-parallel cross-shell readings are

resolved radially. The formal one-paragraph statement is registered for the follow-up script's note companion; until then the two-shell row of this recheck is T3.

- (γ) *“Pre-registering the decagonal as the stress test could bias future verification toward a pattern already known to be energetically weak (zero 4-wave resonances), creating false comfort.”* VALID-with-mitigation: the stress-test role is for the BEYOND-LAYER machinery (packing/energy-bound extremality), where resonance count is irrelevant; for the LAYER-energetics stress test the chain's existing worst cases (BCC's 144 resonant excess; the band interval of Prop A) remain the designated probes. The note records both roles explicitly so neither substitutes for the other.

6. Result footer

Result ID:	B1-RH-ENUM / enumeration-amended-class-recheck (unit U8)
Precise statement:	CROSS-CHECK (T4 on exact geometry): every enumerated/gallery pattern satisfies the amended-class separation $\geq \theta_{\min}$ at all anchor intensities (min margins: decagonal x1.042 anchor / x1.0018 endpoint; all others $\geq$ x1.67). Races and the U1 class theorem compose with NO coverage gap. Decagonal QC = extremal in-class pattern, PRE-REGISTERED as the beyond-layer stress test. k-fold planar stars below the cutoff are sea-reclassified.
Scope:	Single-shell enumerated set + Math437 gallery; two-shell ensembles inherit per-shell (the cross-shell radial-resolution paragraph is T3, registered).
Dependencies:	H-ADM-COH (verdict #14), AddB/AddC scales ($\theta_{\min}$, n_pack), Math437 gallery, U1 entry package.
Evidence grade:	ANALYTIC (exact geometry) + closed-form arithmetic on certified constants; machine asserts NOT executed (registered).
Reproduction command:	(registered follow-up) angle-table asserts in the shared U4/U7 script or a sibling under codes/vacuum/.
Expected output:	exact angles to 1e-12; $\theta_{\min}(I)$ to 1e-6; margin table; JSON under claims/B1-RH-ENUM/runs/.
Falsification gate:	Any enumerated pattern measuring BELOW $\theta_{\min}$ at a certified intensity (would break the composition and demote this note); the decagonal stress test exceeding the STEP-5B envelope (would fire the gate-level falsifier).
Tier before / after:	No tier action; B1 stays T5 (T6 proposal of

U1 unchanged, pending operator).

No-overclaim statement: The recheck certifies CLASS MEMBERSHIP of the enumerated set, not any new energetic statement. The two-shell cross-shell paragraph is T3 until written. PDF build + linter PENDING operator-side.

Next required action: Machine asserts for the angle table; the cross-shell radial-resolution paragraph; mainline continues (U9).